# Return-Conservative Distributional Jump Bridges for Financial Disclosure Modeling

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**Abstract**

Financial disclosure models often let event text directly alter every predicted market-response target, including abnormal return. This is a flexible design, but it can destroy the weaker cross-sectional ranking signal carried by pre-event market state. We propose the Return-Conservative Distributional Jump Bridge (RC-DJB), a neural architecture that first estimates a no-event response distribution and then lets SEC filing text transport volatility, volume, and predictive uncertainty while imposing a hard firewall on the abnormal-return mean. The constraint is falsifiable: if disclosure text contains robust near-term return-ranking signal, blocking its direct return-mean path should hurt; if not, the bridge should improve event-response modeling without overwriting rank information. On a leakage-controlled public-data study with 7,236 SEC 8-K events, 7,236 matched controls, and 2,463 held-out rows from 98 large U.S. firms, RC-DJB obtains abnormal return rank IC 0.0546 with 95% ticker-clustered interval [0.0104, 0.0948]. It is the only tested model that both beats concat fusion on multi-target MSE by a positive paired interval (0.0043, [0.0023, 0.0061]) and has a strictly positive ticker-clustered rank-IC interval. Event bridge interventions increase held-out event-row MSE by 0.0066, 0.0049, and 0.0081 when the bridge is removed, text is zeroed, or text is shuffled. These findings support a return-conservative modeling principle for disclosure events: use filing text to transport risk, liquidity, and uncertainty responses while preserving abnormal-return ranking in the no-event market state.

# 1 Introduction

Public disclosures are timed interventions in the market information set [Fama, 1970, Beaver, 1968, Patell and Wolfson, 1984]. A filing can change volatility, liquidity, and uncertainty even when its effect on cross-sectional abnormal-return ranking is weak or unstable [Bamber, 1986, Kothari, 2001]. Standard multimodal forecasters typically concatenate price history and text embeddings, then predict all targets with a shared head [Vaswani et al., 2017, Lim et al., 2021, Chen et al., 2025]. This design asks the same text representation to explain risk response, volume response, uncertainty, and return ranking. In financial data, that coupling is a liability: return labels are noisy, text is high dimensional, and small leakage, data snooping, or backtest overfitting can look like alpha [Campbell et al., 1997, Gu et al., 2020, White, 2000, Bailey et al., 2014, López de Prado, 2018].

This paper studies a stricter architectural hypothesis. SEC disclosure text should be allowed to transport the event-response distribution, but it should not directly overwrite the mean abnormal return predicted from the no-event market state. We call the resulting model a Return-Conservative Distributional Jump Bridge (RC-DJB). The model first predicts a no-event Gaussian response distribution from pre-event price, volume, and market-relative features. It then uses disclosure text to generate bounded shifts in volatility-jump mean, volume-jump mean, and predictive log-variance. The abnormal-return mean shift is constrained to zero.

The contributions are:

1. A return-conservative distributional bridge architecture for event-driven financial modeling.
2. A leakage-controlled SEC 8-K experiment with matched no-event controls and public post-filing market outcomes.

3. Evidence that blocking direct text-to-return-mean transport recovers statistically positive rank IC while preserving response-regression gains over fusion baselines.

4. Intervention tests showing that the disclosure bridge contributes to event-row MSE even though return ranking is intentionally anchored to the no-event state.

# 2 Related Work

Classical event studies estimate abnormal returns around information releases [Fama et al., 1969, Brown and Warner, 1985, MacKinlay, 1997]. Disclosure and earnings-announcement studies document price, volatility, and trading-volume responses after public information releases [Beaver, 1968, Patell and Wolfson, 1984, Bamber, 1986, Boehmer et al., 1991, Kothari and Warner, 2007]. RC-DJB keeps this information boundary but replaces scalar event-window estimation with a neural distributional response model. The abnormal-return construction follows the market-model and factor-model tradition [Sharpe, 1964, Fama and French, 1993, Carhart, 1997, Kothari and Warner, 2007]; this study uses a SPY-adjusted public-data variant rather than CRSP/factor residuals.

The architecture is related to recurrent, state-space, and dynamical-system methods, including LSTMs and gated recurrent units [Hochreiter and Schmidhuber, 1997, Cho et al., 2014], Koopman operators [Koopman, 1931], neural differential equations [Chen et al., 2018], structured state-space sequence models [Gu et al., 2022], and selective state-space models [Gu and Dao, 2023, Wang et al., 2025]. It also draws on probabilistic forecasting and time-series foundation model work [Dawid, 1984, Gneiting and Raftery, 2007, Salinas et al., 2020, Lim et al., 2021, Nie et al., 2023, Wu et al., 2023a, Ansari et al., 2024, Wu et al., 2024, Jin et al., 2023]. Marked point processes model event arrivals [Hawkes, 1971, Mei and Eisner, 2017]; here the disclosure time is observed and the task is post-event response prediction.

Financial text contains information relevant to prices and fundamentals [Tetlock, 2007, Tetlock et al., 2008, Kogan et al., 2009, Loughran and McDonald, 2011, Cohen et al., 2020]. Recent financial NLP systems and benchmarks study domain language models, multimodal financial forecasting, SEC filing reasoning, and leakage in financial agents [Jegadeesh and Wu, 2013, Araci, 2019, Wu et al., 2023b, Yang et al., 2023, Kong et al., 2025, Chen et al., 2025, Islam et al., 2023, Jiang et al., 2026, Li et al., 2025]. RC-DJB is also adjacent to causal representation and counterfactual outcome models [Rubin, 1974, Rosenbaum and Rubin, 1983, Melnychuk et al., 2022, Feng et al., 2024]. It differs by imposing a target-specific architectural restriction: disclosure text may transport response distribution components, but the abnormal-return mean remains no-event anchored.

# 3 Method

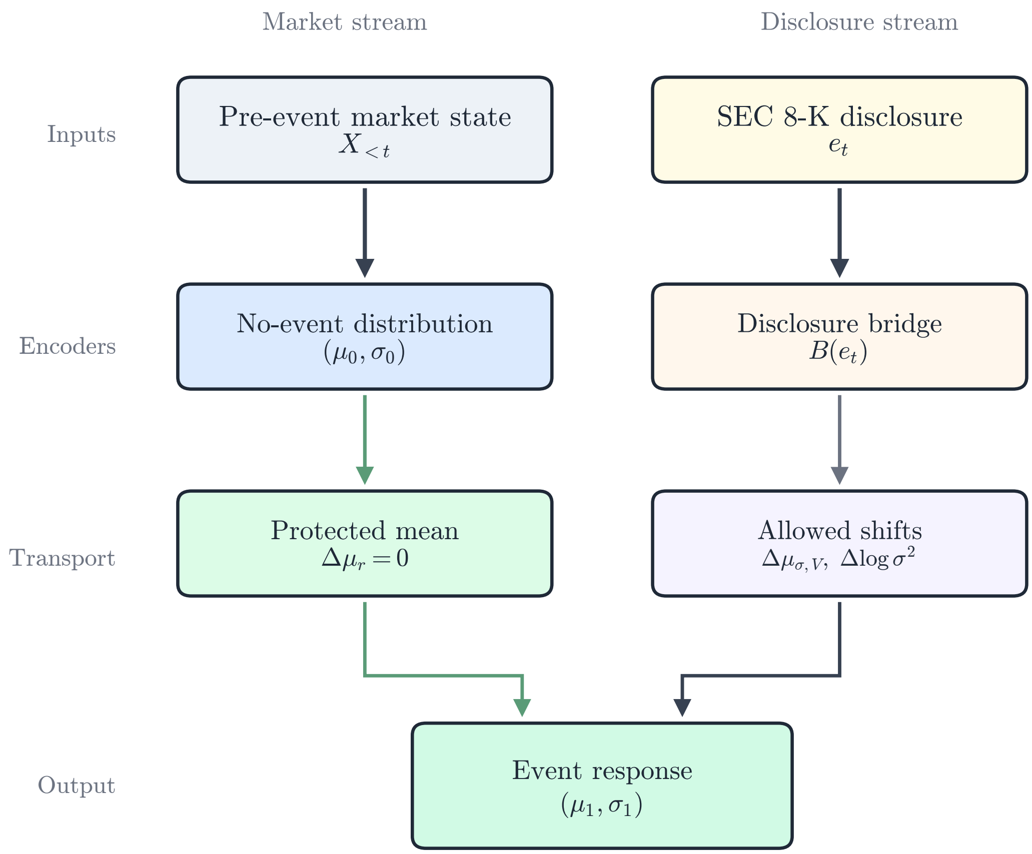


Figure 1: RC-DJB first estimates a no-event response distribution from market history. Disclosure text generates a bounded bridge for risk/liquidity response and predictive uncertainty, but a return-mean firewall enforces $\Delta\mu_r = 0$.

**Inputs and targets.** Each sample $i$ has label start date $\tau_i$, a 40-trading-day pre-event tensor $X_i \in \mathbb{R}^{40\times 8}$, disclosure embedding $e_i$, metadata $m_i \in \mathbb{R}^6$, and target vector $y_i \in \mathbb{R}^3$. The targets are five-trading-day SPY-adjusted return, realized-volatility jump, and volume jump. The headline variant uses chunk-pooled BGE-small disclosure embeddings, following the dense sentence and passage embedding literature [Reimers and Gurevych, 2019, Karpukhin et al., 2020, Xiao et al., 2023].

**Event-study labels.** Let $P_{i,t}$, $V_{i,t}$, and $r_{i,t}$ denote adjusted close, volume, and daily close return for firm $i$, and let $r_t^M$ denote the SPY return. The label window $\mathcal{W}_i = \{\tau_i, \ldots, \eta_i\}$ starts on the disclosure date for before-close traded-day filings and on the next traded date otherwise; $\eta_i$ is the fifth tradable session in the window. The pre-window $\mathcal{P}_i$ is the 20 traded sessions strictly before $\tau_i$. These choices follow the event-study convention that features must precede the public event time while outcomes are measured only after the event window begins [Brown and Warner, 1985, MacKinlay, 1997, Patell and Wolfson, 1984, Boehmer et al., 1991]. Realized-volatility differences are reported as a simple daily-data analogue of the realized-variation literature [Andersen et al., 2003]. The implemented targets are

$$y_{i,r} = \frac{P_{i,\eta_i}}{P_{i,\tau_i}} - 1 - \left(\prod_{t\in\mathcal{W}_i}(1+r_t^M) - 1\right), \tag{1}$$

$$y_{i,\sigma} = \mathrm{sd}_{t\in\mathcal{W}_i}(r_{i,t}) - \mathrm{sd}_{t\in\mathcal{P}_i}(r_{i,t}), \tag{2}$$

$$y_{i,V} = \frac{|\mathcal{P}_i|}{|\mathcal{W}_i|}\frac{\sum_{t\in\mathcal{W}_i} V_{i,t}}{\sum_{t\in\mathcal{P}_i} V_{i,t}} - 1. \tag{3}$$

Rows are admissible only when their maximum feature date is strictly before the label start, $\max\{t : t \in X_i\} < \tau_i$.

**No-event distribution.** A recurrent encoder maps pre-event market history to a latent state [Cho et al., 2014]:

$$h_i = \mathrm{GRU}_\theta(X_i) + W_m m_i. \tag{4}$$

A bounded gate produces the no-event state $\tilde{s}_i$, from which the model predicts a diagonal Gaussian distribution, a standard proper-scoring-rule choice for probabilistic forecasts [Gneiting and Raftery, 2007, Salinas et al., 2020]:

$$\tilde{s}_i = h_i \odot (1 + 0.1\tanh(g_\theta(h_i, m_i))), \tag{5}$$

$$[\mu_{0,i}, \ell_{0,i}] = f_\theta(\tilde{s}_i, m_i), \tag{6}$$

where $\ell_{0,i}$ is log-variance.

**Distributional bridge.** Disclosure text generates a bounded latent state jump and distributional transport:

$$b_i = B_\theta(e_i), \tag{7}$$

$$j_i = \alpha\tanh(u_\theta(b_i)) \odot \sigma(v_\theta(b_i)), \tag{8}$$

$$[\Delta\mu_i, \Delta\ell_i] = T_\theta(\mathrm{LN}(\tilde{s}_i + j_i), b_i, m_i). \tag{9}$$

Here LN denotes layer normalization [Ba et al., 2016]. The central architectural constraint is $\Delta\mu_{i,return} = 0$. Thus,

$$\mu_{1,i} = \mu_{0,i} + [0, \Delta\mu_{i,volatility}, \Delta\mu_{i,volume}], \qquad \ell_{1,i} = \mathrm{clip}(\ell_{0,i} + \Delta\ell_i). \tag{10}$$

The model predicts $p_\theta(y_i) = \mathcal{N}(\mu_{1,i}, \mathrm{diag}(\exp(\ell_{1,i})))$.

**Objective.** For target dimension $k$, the Gaussian negative log likelihood used for probabilistic training and evaluation is a proper likelihood-based scoring objective [Gneiting and Raftery, 2007]:

$$\mathcal{L}_{NLL} = \frac{1}{2NK}\sum_{i=1}^{N}\sum_{k=1}^{K}\left[\exp(-\ell_{1,ik})(y_{ik}-\mu_{1,ik})^2 + \ell_{1,ik}\right], \qquad K=3, \quad \ell_{1,ik}\in[-7,3]. \tag{11}$$

The pairwise rank regularizer is computed only over real-disclosure pairs in a minibatch, following pairwise learning-to-rank objectives [Burges et al., 2005, Cao et al., 2007]:

$$\mathcal{L}_{rank} = \frac{1}{|\mathcal{Q}|}\sum_{(i,j)\in\mathcal{Q}} \log\left(1+\exp\left[-\operatorname{sign}(y_{i,r}-y_{j,r})\frac{\mu_{1,i,r}-\mu_{1,j,r}}{T}\right]\right), \tag{12}$$

where $\mathcal{Q}$ contains event pairs with unequal abnormal-return labels and $T = 0.02$. The supervised term is the weighted Smooth-L1 loss after optional target standardization [Huber, 1964]:

$$\mathcal{L}_{Huber} = \frac{1}{NK}\sum_{i=1}^{N}\sum_{k=1}^{K} w_k \operatorname{SmoothL1}_{\beta}\left(\frac{\mu_{1,ik}-\bar{y}_k}{s_k}, \frac{y_{ik}-\bar{y}_k}{s_k}\right). \tag{13}$$

For bottleneck models with posterior parameters $(\mu_i^z, \ell_i^z)$, the KL penalty follows variational and information-bottleneck regularization [Kingma and Welling, 2013, Tishby et al., 2000],

$$\mathcal{L}_{KL} = -\frac{1}{2}\operatorname{mean}_{i,k}\left(1+\ell_{ik}^z-(\mu_{ik}^z)^2-\exp(\ell_{ik}^z)\right). \tag{14}$$

The control and consistency penalties are

$$\mathcal{L}_{control} = \frac{\sum_i \mathbb{1}\{a_i=0\}\|\Delta\mu_i\|_2^2}{\sum_i \mathbb{1}\{a_i=0\}}, \tag{15}$$

$$\mathcal{L}_{consistency} = \frac{1}{K}\sum_{k=1}^{K}\operatorname{Var}_i(\Delta\mu_{i,k}), \tag{16}$$

where $a_i = 1$ for real disclosures and $a_i = 0$ for matched controls. The full training loss combines target-standardized Huber regression, Gaussian likelihood, control suppression, bottleneck regularization, bridge sparsity, residual consistency, and rank learning:

$$\mathcal{L} = \lambda_y\mathcal{L}_{Huber} + \lambda_n\mathcal{L}_{NLL} + \lambda_c\mathcal{L}_{control} + \lambda_{KL}\mathcal{L}_{KL} + \lambda_s\|z_i\|_1 + \lambda_v\mathcal{L}_{consistency} + \lambda_r\mathcal{L}_{rank}. \tag{17}$$

Here $z_i$ is the stored disclosure-bridge latent used for sparsity diagnostics. The main run uses $w_y = (3.0, 1.0, 0.25)$, $\lambda_n = 0.10$, $\lambda_c = 0.35$, $\lambda_{KL} = 0.01$, $\lambda_s = 0.001$, $\lambda_v = 0.05$, and $\lambda_r = 0.15$.

# 4 Data and Leakage Controls

The experiment uses public SEC EDGAR APIs [U.S. Securities and Exchange Commission, 2026]. The sample covers 98 usable large U.S. firms, SEC 8-K filings from 2020–2025, and public daily prices. Tickers are mapped to CIKs through the SEC company-tickers file. Filings are read from SEC submissions JSON, including archived submissions, using `acceptanceDateTime` and the primary EDGAR document URL. The parser stores source URL, accession number, accepted timestamp, available timestamp, extracted text, and text hash. Prior filing-text studies motivate using public SEC text as a market-response substrate [Loughran and McDonald, 2011, Jegadeesh and Wu, 2013, Cohen et al., 2020].

Daily prices are downloaded from public price endpoints, with SPY used as the market proxy for abnormal returns. The trading calendar is inferred from observed price rows. Event label windows start on the same observed trading day for filings accepted before 16:00 ET; filings accepted after 16:00 ET or on non-trading days start on the next observed trading day. Features use only prices strictly before the label start date. Labels use future returns only as targets. One deterministic same-ticker no-event control is constructed for each filing, excluding dates within a 10-calendar-day blackout window around known events. The no-lookahead rule and blackout controls are motivated by standard event-study design and known financial backtesting leakage failure modes [MacKinlay, 1997, White, 2000, Hansen, 2005, Bailey et al., 2014, López de Prado, 2018, Li et al., 2025]. Figure 2 visualizes the chronological sample and the feature-to-label guard margin used by every row.

Table 1: Main public-data sample.

| Artifact | Count |
|---|---:|
| Usable companies | 98 |
| SEC 8-K events | 7,236 |
| Matched no-event controls | 7,236 |
| Feature rows | 14,472 |
| Public price rows | 159,093 |
| Held-out rows | 2,463 |

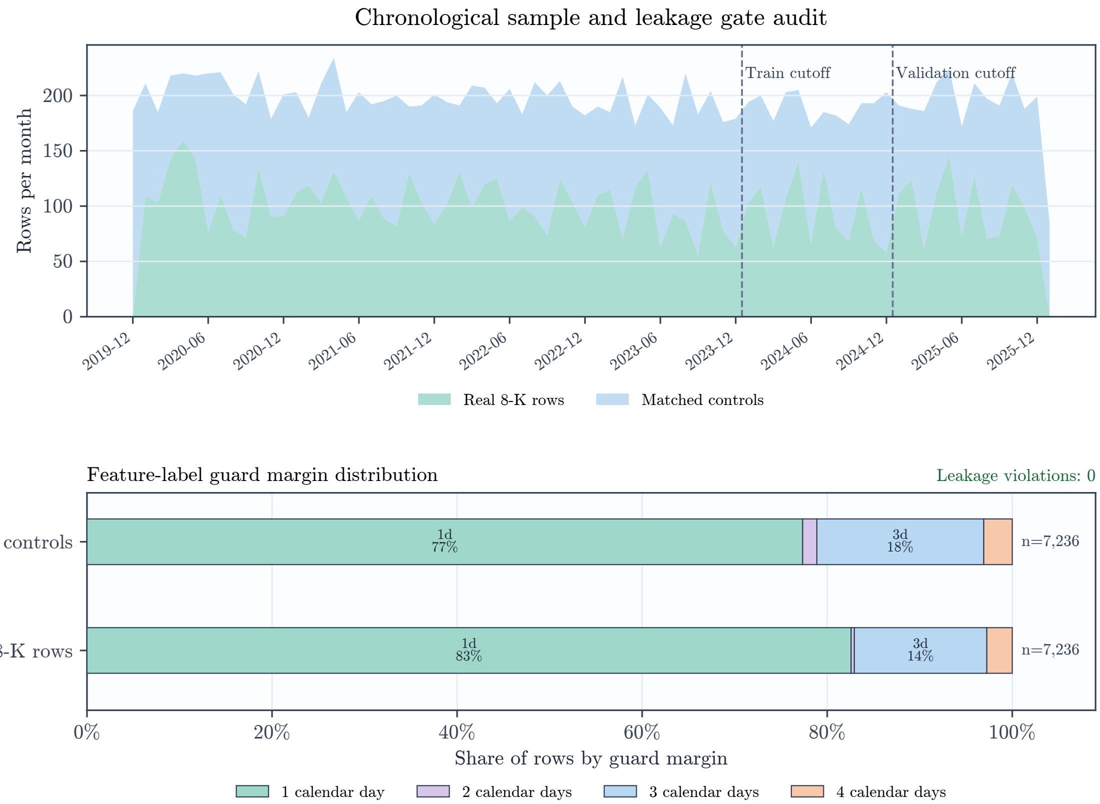


Figure 2: Chronological data substrate and feature-label leakage audit. The upper panel shows monthly real-event and matched-control rows with chronological split boundaries; the lower panel shows the share of rows by positive feature-label guard margin and the zero-violation leakage audit.

# 5 Experiments

Rows with $\tau_i \leq$ 2023-12-31 are training rows, rows from 2024 are validation rows, and later rows are held out. This yields 9,729 training rows, 2,280 validation rows, and 2,463 held-out rows across 98 held-out tickers and 13 held-out label-start months. The headline RC-DJB checkpoint is selected by validation loss. All models use fixed seed 29 and are evaluated once on the held-out split. The chronological split follows time-series evaluation practice, where future observations must not influence model selection or feature construction [Bergmeir and Benítez, 2012, Hyndman and Athanasopoulos, 2021, López de Prado, 2018].

Baselines include no-event dynamics, text-only MLP, concat fusion, Counterfactual Event Bottleneck Transformer (CEBT), Disclosure Operator Transformer (DOT), Event-Jump State Space Model (EJSSM), and an unconstrained Distributional Jump Bridge (DJB). Confidence intervals use percentile bootstrap resampling; ticker-clustered intervals resample tickers with replacement [Efron, 1979, Efron and Tibshirani, 1993, Davison and Hinkley, 1997, Cameron et al., 2008, Cameron and Miller, 2015]. Paired comparisons use row-paired bootstrap resampling. Rank IC is Spearman correlation between predicted and realized abnormal return [Spearman, 1904, Grinold and Kahn, 2000].

**Evaluation statistics.** For model $m$, the reported multi-target error is

$$\mathrm{MSE}(m) = \frac{1}{NK}\sum_{i=1}^{N}\sum_{k=1}^{K}\left(y_{ik} - \hat{y}_{ik}^{(m)}\right)^2. \tag{18}$$

Rank IC is the Spearman correlation between predicted and realized abnormal return:

$$\mathrm{IC}(m) = \rho_P\left(\mathrm{rank}(\hat{y}_{1:N,r}^{(m)}), \mathrm{rank}(y_{1:N,r})\right). \tag{19}$$

For a statistic $S$, bootstrap replicates $S^{*(1)}, \ldots, S^{*(B)}$ form percentile intervals

$$\mathrm{CI}_{0.95}(S) = \left[Q_{0.025}\{S^{*(b)}\}_{b=1}^{B}, Q_{0.975}\{S^{*(b)}\}_{b=1}^{B}\right], \tag{20}$$

The model-level intervals in Table 2 use $B = 1000$. Ticker-clustered intervals sample tickers with replacement and include all held-out rows for each sampled ticker. Rank IC uses integer ranks of predicted and realized abnormal returns. Paired MSE comparisons against RC-DJB use $B = 2000$ row-paired bootstrap replicates and

$$\Delta_{\mathrm{MSE}}(m) = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{1}{K}\|y_i - \hat{y}_i^{(m)}\|_2^2 - \frac{1}{K}\|y_i - \hat{y}_i^{RC}\|_2^2\right], \tag{21}$$

with rows resampled as matched pairs. Paired rank comparisons use the same joined rows but report $\mathrm{IC}(RC) - \mathrm{IC}(m)$.

**Probabilistic diagnostics.** For nominal level $1-\alpha$, the diagonal Gaussian predictive interval for target $k$ is

$$I_{ik}(1-\alpha) = \left[\mu_{1,ik} - z_{1-\alpha/2}\exp(\ell_{1,ik}/2), \mu_{1,ik} + z_{1-\alpha/2}\exp(\ell_{1,ik}/2)\right], \tag{22}$$

and empirical coverage is

$$\widehat{C}(1-\alpha) = \frac{1}{NK}\sum_{i=1}^{N}\sum_{k=1}^{K}⊮\{y_{ik} \in I_{ik}(1-\alpha)\}. \tag{23}$$

Mean predictive standard deviation is $\frac{1}{NK}\sum_{i,k}\exp(\ell_{1,ik}/2)$. Interval coverage is reported because calibrated predictive uncertainty is distinct from point forecast error [Winkler, 1972, Gneiting and Raftery, 2007, Gneiting et al., 2007, Gneiting and Katzfuss, 2014].

Table 2: Held-out results. MSE is over abnormal return, volatility jump, and volume jump. Intervals are 95% ticker-clustered bootstrap intervals.

| Model | MSE | MSE CI | Rank IC | Rank IC CI |
|---|---|---|---|---|
| No-event | 0.0696 | [0.0519, 0.0936] | -0.0411 | [-0.0840, 0.0016] |
| Concat fusion | 0.0597 | [0.0442, 0.0817] | -0.0182 | [-0.0597, 0.0267] |
| CEBT | 0.0660 | [0.0488, 0.0898] | 0.0463 | [0.0052, 0.0893] |
| DOT | 0.0596 | [0.0449, 0.0804] | -0.0131 | [-0.0510, 0.0291] |
| EJSSM-BGE | 0.0547 | [0.0395, 0.0764] | 0.0099 | [-0.0329, 0.0556] |
| DJB | **0.0539** | [0.0389, 0.0757] | 0.0128 | [-0.0290, 0.0547] |
| RC-DJB | 0.0554 | [0.0400, 0.0776] | **0.0546** | [0.0104, 0.0948] |

Table 3: Paired comparisons against RC-DJB. Positive MSE values mean RC-DJB has lower MSE. Positive rank values mean RC-DJB has higher rank IC.

| Baseline | MSE Diff. | 95% CI | Rank IC Diff. | 95% CI |
|---|---|---|---|---|
| No-event | 0.01417 | [0.01150, 0.01747] | 0.0956 | [0.0412, 0.1504] |
| Concat fusion | 0.00427 | [0.00230, 0.00612] | 0.0728 | [0.0197, 0.1249] |
| CEBT | 0.01057 | [0.00825, 0.01311] | 0.0083 | [-0.0500, 0.0668] |
| DOT | 0.00418 | [0.00180, 0.00638] | 0.0677 | [0.0186, 0.1180] |
| EJSSM-BGE | -0.00076 | [-0.00163, 0.00005] | 0.0447 | [-0.0072, 0.0952] |

**Bridge diagnostics.** The response-transport statistic for group $c \in \{0, 1\}$ is

$$T_c = \mathrm{mean}_{i:a_i=c,k} \left|\Delta\mu_{i,k}\right|, \tag{24}$$

where $c = 1$ denotes real 8-K rows and $c = 0$ denotes controls. Latent bridge AUC uses the score $s_i = \mathrm{mean}_k |z_{i,k}|$ to classify real disclosures against matched controls, with average ranks used for tied scores [Hanley and McNeil, 1982].

Table 4: RC-DJB diagnostics on the held-out split.

| Diagnostic | Value |
|---|---|
| Gaussian NLL | -2.2864 |
| Observed 80% interval coverage | 0.9011 |
| Observed 95% interval coverage | 0.9602 |
| Mean predictive standard deviation | 0.1302 |
| Mean response transport, real 8-Ks | 0.0358 |
| Mean response transport, controls | 0.0073 |
| Latent bridge AUC, events vs. controls | 0.6038 |

**Intervention definitions.** The no-bridge intervention is the implementation's no-jump forward pass: it sets the bridge state jump, bridge hidden state, mean delta, and log-variance delta to zero, so $\mu_1 = \mu_0$ and $\ell_1 = \ell_0$. The zero-text intervention replaces $e_i$ with the zero vector before the forward pass. The shuffled-text intervention applies one deterministic held-out permutation of disclosure embeddings. For RC-DJB, $\Delta\mu_{return} = 0$ in every variant, so abnormal-return point predictions and rank IC are invariant to these bridge interventions. These ablations are intervention-style tests of a learned component rather than causal identification of market effects [Rubin, 1974, Imbens and Rubin, 2015].

Table 5: RC-DJB intervention tests. Rank IC is unchanged by bridge interventions because the return-mean firewall fixes the abnormal-return point prediction to the no-event mean.

| Variant | All MSE | Event MSE | Event NLL | Rank IC | Bridge AUC |
|---|---|---|---|---|---|
| Full RC-DJB | **0.0554** | **0.0843** | **-2.2115** | 0.0546 | 0.6038 |
| No bridge | 0.0586 | 0.0909 | -2.1204 | 0.0546 | 0.5000 |
| Zero text | 0.0578 | 0.0892 | -2.1886 | 0.0546 | 0.4843 |
| Shuffled text | 0.0597 | 0.0924 | -2.1765 | 0.0546 | 0.5057 |

Table 6: Paired event-row MSE stress tests. Positive values mean full RC-DJB has lower event-row MSE than the intervention.

| Intervention | Event MSE increase | 95% paired CI |
|---|---|---|
| No bridge | 0.0066 | [0.0043, 0.0093] |
| Zero text | 0.0049 | [0.0027, 0.0073] |
| Shuffled text | 0.0081 | [0.0051, 0.0117] |

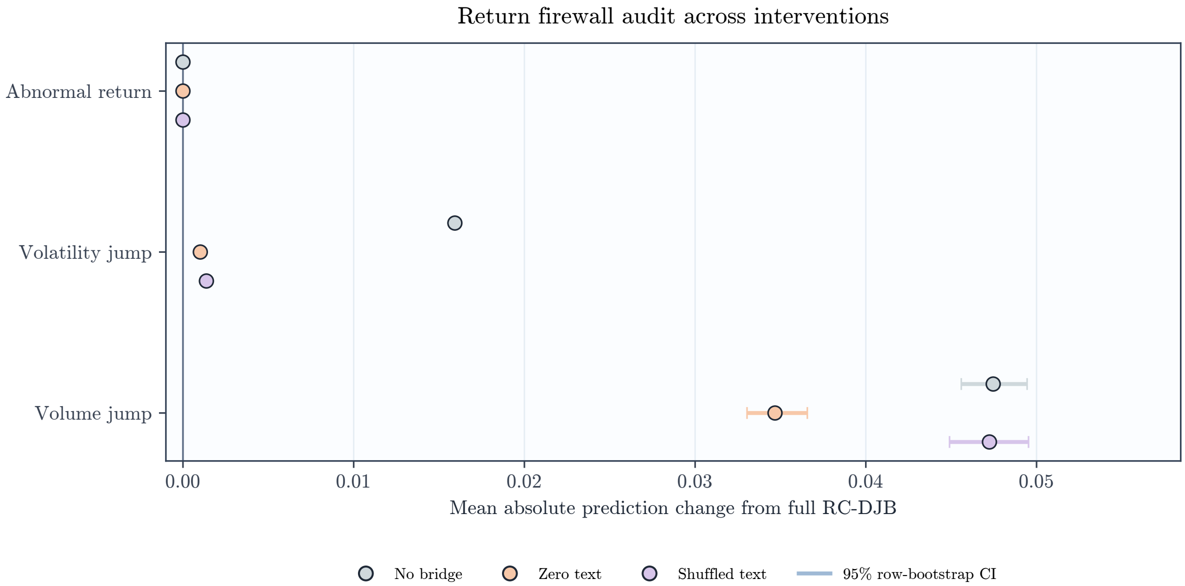


Figure 3: Return firewall audit. Points show mean absolute prediction change from full RC-DJB under bridge interventions, with 95% row-bootstrap intervals. The abnormal-return prediction remains exactly invariant while response targets move.

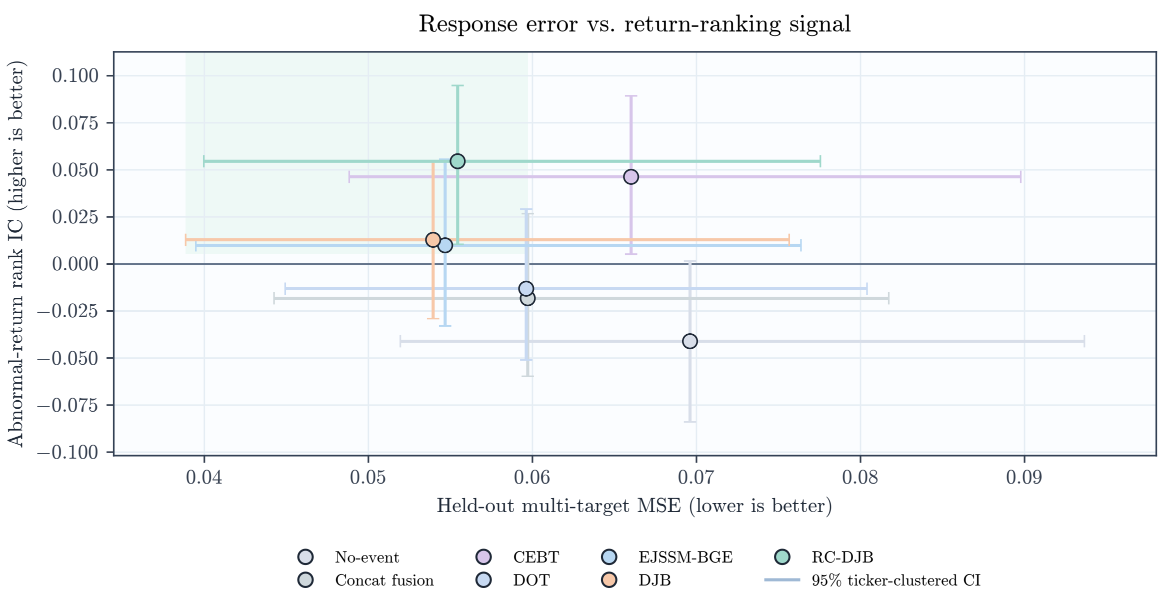


Figure 4: RC-DJB occupies a different point on the response/ranking tradeoff: EJSSM and DJB have lower point MSE, while RC-DJB is the only tested low-error model with a strictly positive ticker-clustered rank-IC interval.

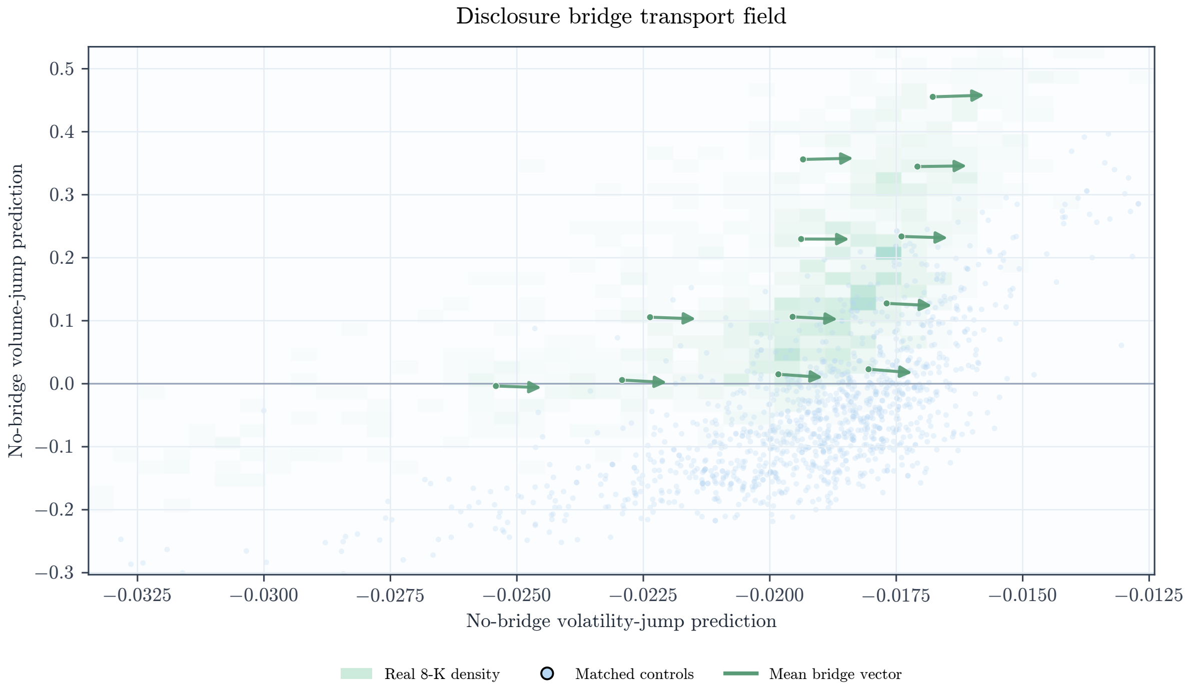


Figure 5: Held-out response-state transport induced by the disclosure bridge. The background shows real 8-K density in the no-bridge volatility/volume response plane, blue points show matched controls, and arrows show binned mean shifts in volatility-jump and volume-jump predictions.

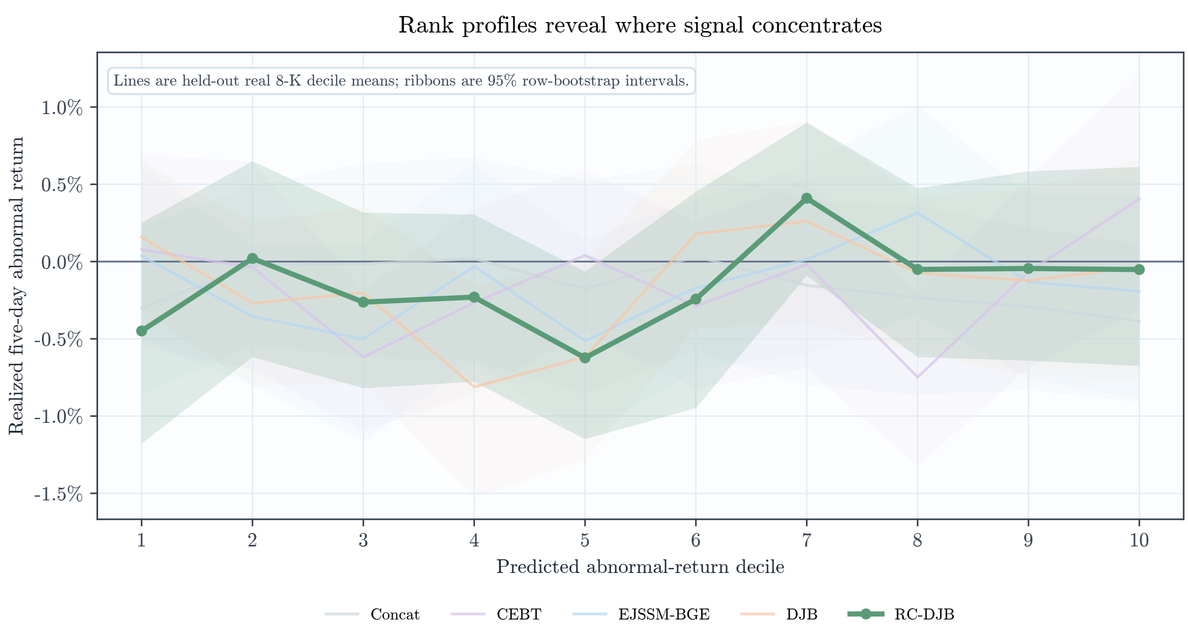


Figure 6: Predicted abnormal-return decile profiles on held-out real 8-K rows. Lines show realized five-day abnormal-return means; ribbons are 95% row-bootstrap confidence intervals computed within each predicted decile.

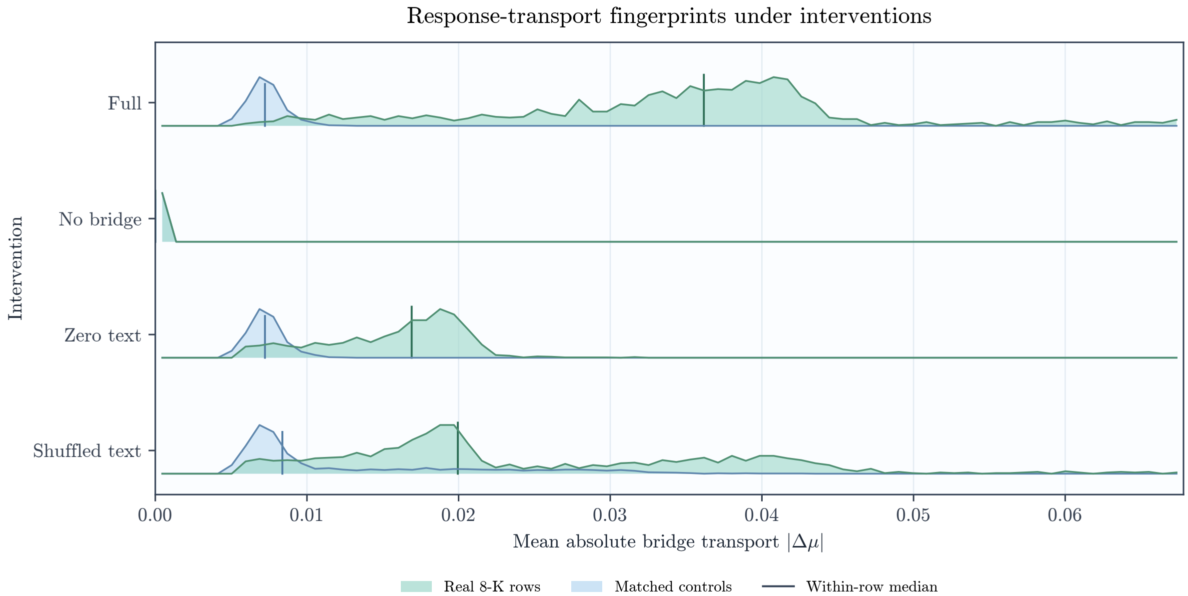


Figure 7: Response-transport fingerprints under bridge interventions. Ridge densities compare mean absolute response transport for real disclosure rows and matched controls; vertical marks show within-row medians.

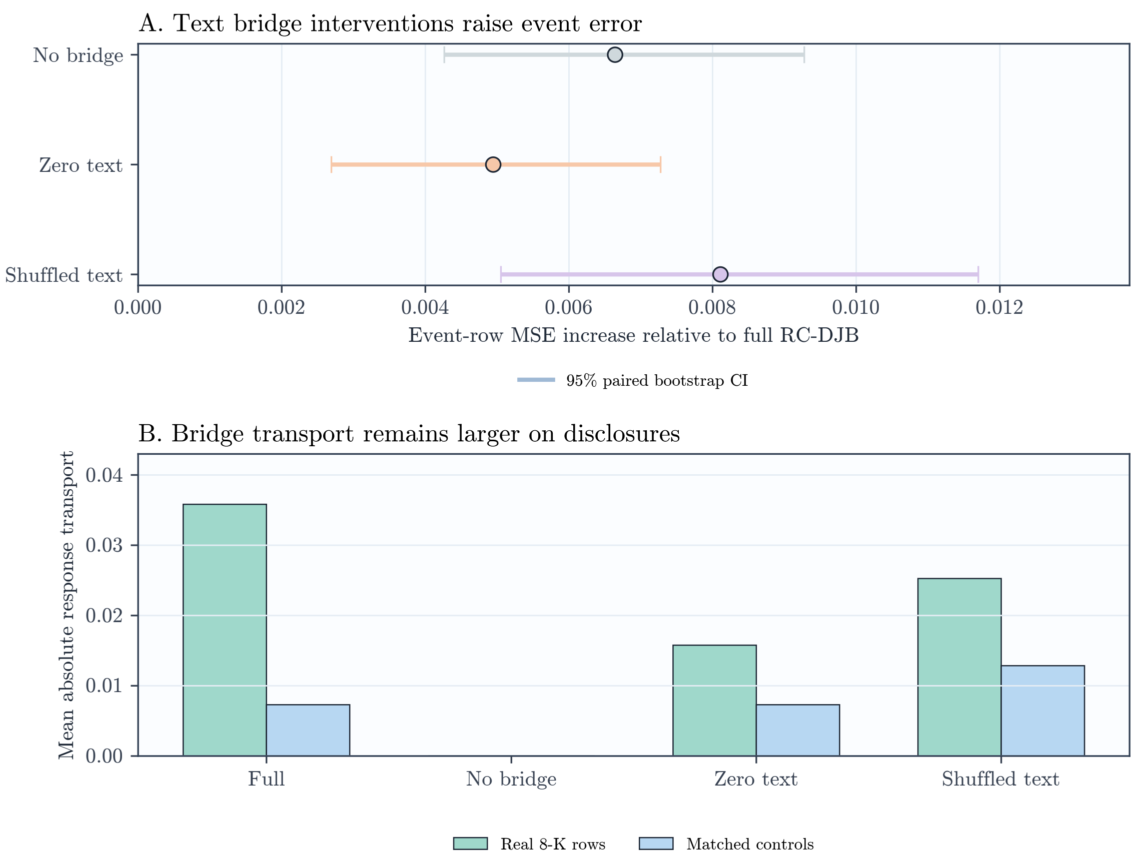


Figure 8: Interventions show that the disclosure bridge matters for event-response error. Rank IC does not move under text interventions because the return-mean firewall prevents direct text-to-return overwriting.

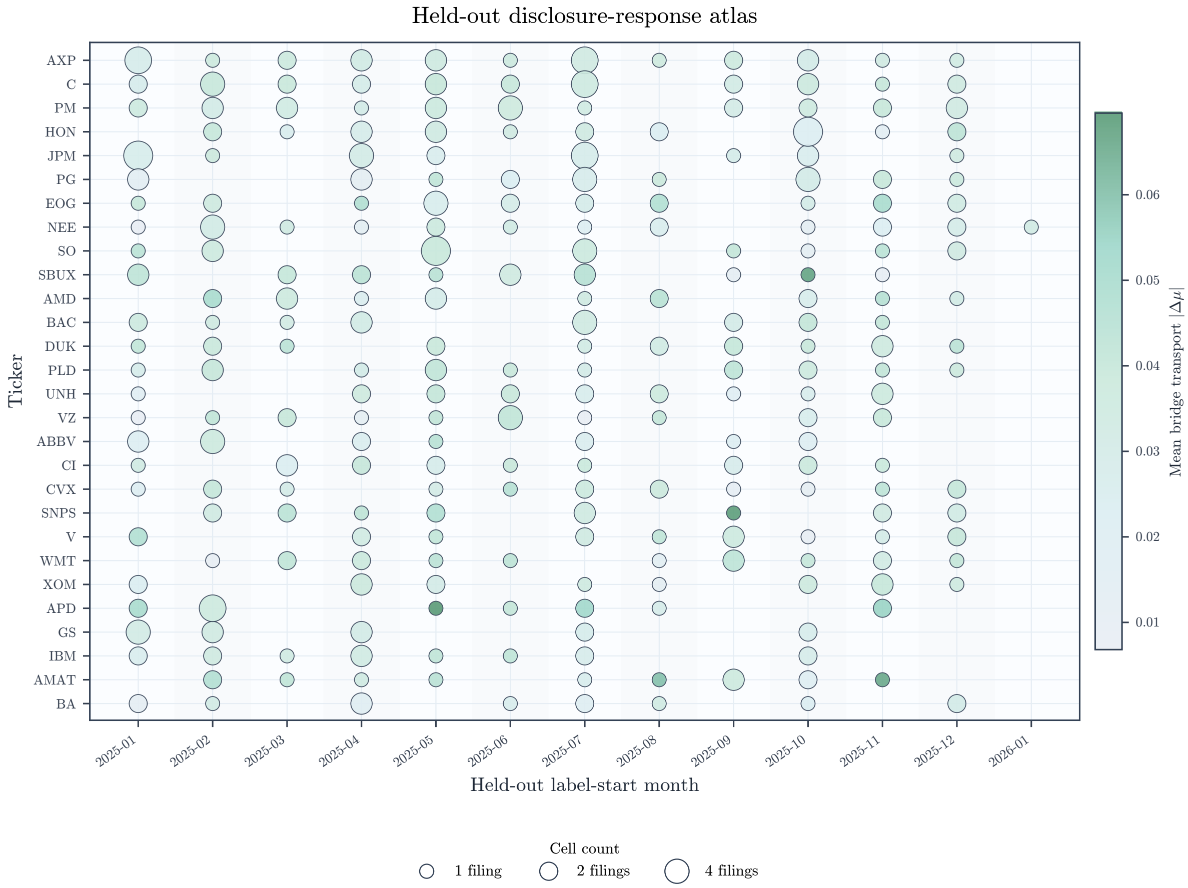


Figure 9: Held-out disclosure-response atlas for the most frequent test tickers. Marker area is the number of filings in the ticker-month cell and marker color is mean bridge transport, exposing heterogeneity across firms and calendar months.

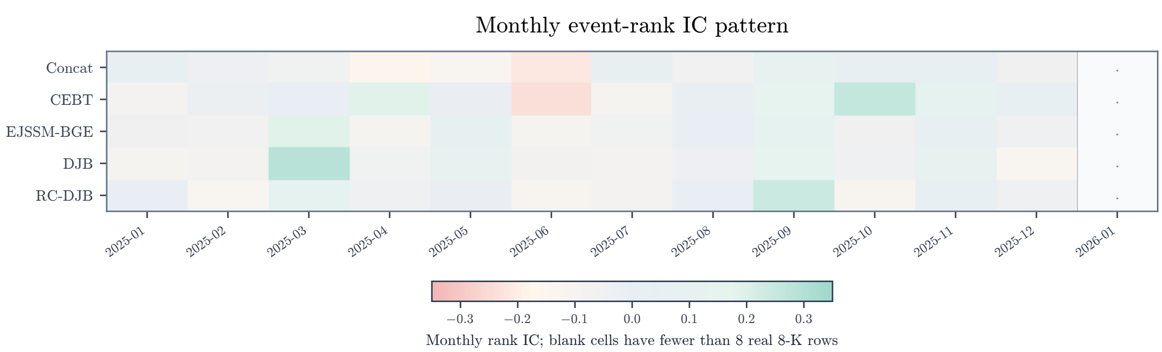


Figure 10: Monthly event-rank IC over held-out real 8-K rows. The model-level rank result is not uniform across months, but RC-DJB has a more positive held-out pattern than fusion and unconstrained event-mean transport baselines.

# 6 Findings

**Return-mean conservation recovers rank signal.** RC-DJB obtains rank IC 0.0546 with ticker-clustered interval [0.0104, 0.0948]. In contrast, concat fusion, DOT, EJSSM-BGE, and unconstrained DJB all have rank intervals crossing zero. CEBT also has positive rank IC, but its MSE is much worse than RC-DJB. The decile profiles in Figure 6 show that this signal is uneven and noisy, rather than a monotone large-effect ranking curve.

**The bridge improves response modeling without moving return means.** RC-DJB improves paired MSE over concat fusion by 0.0043 and over DOT by 0.0042. Removing the bridge raises event-row MSE by 0.0066; zeroing or shuffling text raises event-row MSE by 0.0049 and 0.0081. These changes occur even though abnormal-return predictions are unchanged by construction, which means the bridge is acting through volatility, volume, and uncertainty transport. Figures 5 and 7 visualize the same mechanism: disclosure text changes response-distribution coordinates while matched controls remain closer to zero transport. Figure 3 audits the architectural constraint directly: return predictions are invariant under bridge interventions while volatility and volume predictions move.

**There is a real response/ranking tradeoff.** EJSSM-BGE and unconstrained DJB achieve lower point MSE than RC-DJB, but their rank IC intervals include zero. RC-DJB is therefore not a universal winner; it is a Pareto-frontier point that prioritizes statistically reliable ranking while keeping response error below fusion baselines. This tradeoff is the main empirical evidence for the return firewall.

**The likelihood result is competitive but not decisive.** RC-DJB has Gaussian NLL -2.2864, close to EJSSM-BGE's -2.2839. The paired NLL difference against EJSSM-BGE is small and not statistically decisive. The likelihood evidence supports the bridge as competitive; the distinctive result is the combination of positive rank IC and response-error improvement over fusion baselines.

**The bridge response is heterogeneous.** The ticker-month atlas in Figure 9 shows that bridge transport is not a uniform offset applied to all disclosures. Transport magnitude varies across firms and months, consistent with event-study evidence that disclosure response depends on the event, firm, and market state [Beaver, 1968, Patell and Wolfson, 1984, Kothari and Warner, 2007].

# 7 Reproducibility

The artifact records configurations, seeds, processed event metadata, text hashes, price rows, feature metadata, checkpoints, predictions, table CSVs, and figure outputs used for the reported results. Tests cover timestamp gating, after-hours label starts, feature leakage rejection, control construction, deterministic cached tensors, finite model losses, probabilistic loss integration, and evaluation leakage guards. Labels are computed only from future price windows and are never included in model inputs.

# 8 Limitations

The study has several limitations. The universe is large-cap biased and may contain survivorship effects, ticker-history issues, and 8-K item-mix sensitivity. The price source is public daily data rather than CRSP or intraday trades and quotes. Abnormal returns are SPY-adjusted rather than factor-model residuals [Fama and French, 1993, Carhart, 1997]. BGE-small is a general-purpose encoder, and filings are represented by chunk mean pooling rather than a trained long-document financial encoder. Rank IC is statistically positive but economically small; transaction costs, capacity, and portfolio construction are outside the scope of this study [Grinold and Kahn, 2000, Novy-Marx and Velikov, 2016, Frazzini et al., 2018, López de Prado, 2018]. The return firewall is an architectural and predictive constraint, not structural causal identification [Rubin, 1974, Imbens and Rubin, 2015].

# 9 Ethics and Use

This paper studies event representations using public filings and public daily prices. The results do not constitute trading recommendations. The artifact is intended to support leakage auditing, independent reproduction, and falsification of the empirical claims.

# 10 Conclusion

RC-DJB shows that the route by which disclosure text enters a financial event model matters. Letting text directly move the abnormal-return mean is not necessary for response modeling and appears to harm ranking in this setting. A return-conservative distributional bridge obtains positive held-out rank IC while still improving multi-target event-response MSE over concat fusion and DOT. The evidence supports a more precise architecture principle for disclosure modeling: use text to transport response risk, liquidity, and uncertainty, but keep near-term return ranking anchored to the no-event market state unless stronger evidence justifies crossing that boundary.

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